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Roku WebDriver

The Roku WebDriver is required to control an app. It can be used in conjunction with the Roku Robot Framework Library (/docs/developer-program/dev-tools/automated-channel-testing/robot-framework-library.md), Roku JavaScript library (/docs/developer-program/dev-tools/automated-channel-testing/javascript-library.md), another test framework, or a programming language or a programming language such as Python, Java, or Go to execute test cases.

Roku WebDriver APIs

Roku's WebDriver includes a set of Selenium-based REST APIs for sending commands to a Roku device. These APIs conform to the WebDriver standards (https://github.com/SeleniumHQ/selenium/wiki/JsonWireProtocol) specified by the World Wide Web Consortium (W3C). Specifically, the Roku WebDriver provides an HTTP-compliant JSON wire protocol with endpoints that map to their respective commands.

Path segments that are prefixed with a colon (:) represent variables. For example, the `:sessionId` variable is included in most command paths. This variable represents the ID of the session to be retrieved or the session where a command is to be sent.

The following table lists the available commands:

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HTTP Method	Path	Summary
GET	/status	Queries the server's current status.
POST	v1/session	Creates a new session.
GET	v1/sessions	Returns a list of the currently active sessions.
GET	v1/session/:sessionId	Retrieves information about the specified session.
DELETE	v1/session/:sessionId	Deletes the session.
POST	v1/session/:sessionId/input (*available since release 2.0*)	Deep links into content while the app is already running.
POST	v1/session/:sessionId/install	Installs the specified app.
POST	v1/session/:sessionId/launch	Launches the specified app.
POST	v1/session/:sessionId/load (*available since release 2.0*)	Sideloads the specified app.
POST	v1/session/:sessionId/press	Simulates a keypress on a Roku remote control.
POST	v1/session/:sessionId/timeouts	Configures the amount of time that a specific operation can be executed before it is aborted.
POST	v1/session/:sessionId/timeouts/press_wait	Configures the amount of time between press cmd execution (if a button_sequence is used in the /press endpoint)
POST	v1/session/:sessionId/timeouts/implicit_wait	Configures the amount of time that a command can be executed before it is aborted.
POST	v1/session/:sessionId/element	Searches for an element on the screen.
POST	v1/session/:sessionId/elements	Searches for multiple elements on the page, starting from the screen root.
POST	v1/session/:sessionId/element/active	Gets the element on the page that currently has focus.

HTTP Method	Path	Summary
GET	v1/session/:sessionId/apps	Returns a list of apps installed on the device.
GET	v1/session/:sessionId/current_app	Returns information about the app currently loaded on the device.
GET	v1/session/:sessionId/source	Gets the current screen source.

Command requests

All command requests and POST/PUT message bodies are sent with a content-type of `application/json;charset=UTF-8`.

Command responses

Command responses are sent as HTTP/1.1 response messages (http://www.w3.org/Protocols/rfc2616/rfc2616-sec6.html#sec6). The following sections describe how the successful, invalid, and failed commands responses are sent.

Success

For successful requests, a 2xx HTTP response is returned. Successful command responses and the included message body are sent with a Content-Type of `application/json;charset=UTF-8`. The JSON message body includes the following properties:

Key	Type	Description
sessionId	string	The advertising ID of the device.

Key	Type	Description
status	number	A status code summarizing the result of the command:

Code	Summary	Detail
0	`Success`	The command executed successfully.
6	`NoSuchDriver`	A session is either terminated or not started
7	`NoSuchElement`	An element could not be located on the page using the given search parameters.
9	`UnknownCommand`	The requested resource could not be found, or a request was received using an HTTP method that is not supported by the mapped resource.
13	`UnknownError`	An unknown server-side error occurred while processing the command.
21	`Timeout`	An operation did not complete before its timeout expired.
32	`InvalidSelector`	Argument was an invalid selector.
33	`SessionNotCreatedException`	A new session could not be created.

value	`*`	The response JSON value.

Invalid

For invalid requests (unknown command or resource not found), a 4xx HTTP response is returned. Invalid command responses are sent with a content-type of `text-plain`, and include a message body with a descriptive error message.

Failed

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If a request maps to a valid command and contains all of the expected parameters in the request body, but fails to execute successfully, a 500 Internal Server Error is returned. The response and included message body have a Content-Type of `application/json;charset=UTF-8`. The message body includes two JSON objects—one with the applicable command response status, and the other with a description of the failure:

Key	Type	Description
status	number	A status code summarizing the result of the command. See the success section for the possible values.
message	string	A descriptive message for the command failure.

GET /status

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Method Type	Path	Return Value	Description
GET	status	A JSON object with the server's platform and build date. This object contains the following fields:	Queries the server's current status and returr the general state of the server. A 200 OK response returned if the server is alive anc accepting commands. This method returns The serve should respond with a general "HTTP 200 OK" response if it . Th response body should be a JSON object describing

Key	Type	Description
sessionId	string	The advertising The advertising ID of the device
status	number	The status code summarizing the result of the command.
value	object	
value.build	object	The build element contains the following attributes: *version* and *time*.
value.build.version	string	A generic release label.
value.build.time	string	A timestamp specifying when the server was built.
value.os	object	The os element contains the following attributes: *arch* and *name*.
value.os.arch	string	The current system architecture.
value.os.name	string	The name of the operating system the server is currently running on (for example, "windows", "linux", and so on).

POST v1/session

Method Type	Path	Parameters	Return Value
POST	session	ip - {string}: The IP address of the device. Example: { "ip": "117.1.1.1" }	A JSON object with the device's advertisement ID, which is used as the sessionId. This object has the following fields:

Key	Type	Description
sessionId	string	The advertisement ID of the device.
status	int	A status code summarizing the result of the command.
value	object	
value.vendorName	string	The vendor of the device.
value.modelName	string	The model of the device.
value.language	string	The language of the device.
value.country	string	The country of the device.
value.ip	string	The IP address of the device.
value.timeout	int	The specified timeout for WebDriver client requests.
value.pressDelay	int	The specified delay between key presses.

GET v1/sessions

Method Type	Path	Return Value	Description
GET	sessions	A JSON object with an array of sessions. This object has the following fields: Key | Type | Description sessionId | string | The advertisement ID of the device. status | int | A status code summarizing the result of the command. value | object | value[i].vendorName | string | The vendor of the device. value[i].modelName | string | The model of the device. value[i].language | string | The language of the device. value[i].country | string | The country of the device. value[i].ip | string | The IP address of the device. value[i].timeout | int | The specified timeout for ECP client requests. value[i].pressDelay | int | The specified delay between key presses.	Returns a list of the currently active sessions.

GET v1/session/:sessionId

Method Type	Path	Return Value	Descri
GET	session/:sessionId	A JSON object with device information. This object has the following fields:	Returr inform on the specifi URL p

Key	Type	Description
sessionId	string	The advertisement ID of the device.
status	int	A status code summarizing the result of the command.
value	object	
value.vendorName	string	The vendor of the device.
value.modelName	string	The model of the device.
value.language	string	The language of the device.
value.country	string	The country of the device.
value.ip	string	The IP address of the device.
value.timeout	int	The specified timeout for WebDriver client requests.
value.pressDelay	int	The specified delay between key presses.

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DELETE v1/session/:sessionId

Method Type	Path	Return Value	Description
DELETE	session/:sessionId	A JSON object that has the following fields: Key | Type | Description sessionId | string | The advertising ID of the device. status | int | A status code summarizing the result of the command. value | object | null	Deletes the session specified in the URL path.

POST v1/session/:sessionId/input

(*available since release 2.0*)

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Method Type	Path	JSON Parameters	Return Value
POST	session/:sessionId/input	channelId - {number}: The ID of the app to be launched. contentId - {string} (optional): The contentId (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentType to execute deep linking tests. contentType - {string} (optional): The mediaType (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentId to execute deep linking tests. Example: `{ "channelId": "dev", "contentId": "myMovie123", "contentType": "movie" }`	A JSON object with Key: sessionId, Type: string Key: status, Type: int Key: value, Type: objec

POST v1/session/:sessionId/install

Method Type	Path	JSON Parameters	Return Value

POST	session/:sessionId/install	channelId - {number}: The ID of the app to be installed.
		
		Example:
		
		{
		"channelId": "dev"
		}
			A JSON object with the following
			
			Key: sessionId — Type: string — Descript: The adve of the de
			Key: status — Type: int — Descript: A status summari result of comman
			Key: value — Type: object — Descript: null

POST v1/session/:sessionId/launch

Method Type	Path	JSON Parameters	Return Value

POST	session/:sessionId/launch	channelId - {number}: The ID of the app to be launched.
		
		contentId - {string} (optional): The contentId (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentType to execute deep linking tests.
		
		contentType - {string} (optional): The mediaType (/docs/developer-program/discovery/implementing-deep-linking.md#understanding-deep-linking-parameters) of the content to be played. You can include this parameter and the contentId to execute deep linking tests.
		
		Example:
		
		{
		"channelId": "dev",
		"contentId": "myMovie123",
		"contentType": "movie"
		}
			A JSON object wit
			
			Key: sessionId, status, value
			Typ: stri, int, obje

POST v1/session/:sessionId/load

(*available since release 2.0*)

Method Type	Path	JSON Parameters	Return Value

POST	session/:sessionId/load	channel - {file}: A zipped package file.
		
		username - {file}: Enter rokudev, which is the user name for the Development Application Installer.
		
		password - {file}: The password for accessing the Development Application Installer on your Roku device.
		
		Example:
		
		{
		"channel": "myChannel.zip",
		"username": "rokudev",
		"password": "your_device_password",
		}
			A JSON object w
			
			Key | Ty
			sessionId | st
			status | in
			value | ob

POST v1/session/:sessionId/press

Method Type	Path	JSON Parameters	Return Value
POST	session/:sessionId/press	button - {string}: The name of the key to be pressed ("home", "up", "down", "left", "right"). button_sequence - {array: string}: An array of keys to be pressed in the specified sequence. button_delays - {array: string} (optional): An array of delays (in ms) between buttons executions. The default value is 1000ms. Example: `{ "button_sequence": ["up", "down", "left"], "buttons_delays": ["1000", "2000"] }` In this example, the delay after the "up" keypress is 1000ms and 2000ms after the "down" keypress.	A JSON objec Key sessionId status value

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POST v1/session/:sessionId/timeouts

Method Type	Path	JSON Parameters	Return Value
POST	session/:sessionId/timeouts	type - {string}: Either "implicit" (ECP commands) or "pressDelay" (delay between press cmd execution) ms - {number}: The amount of time, in milliseconds, that time-limited commands are permitted to run. Example: `{ "type": "implicit", "ms": 2000 }`	A JSON object with the specifi This object has the following fi Key / Type / Descrip sessionId / string / The ad of the status / int / A statu summa result comma value / object / null

POST v1/session/:sessionId/timeouts/implicit_wait

Method Type	Path	JSON Parameters	Return Value
POST	session/:sessionId/timeouts/implicit_wait	ms - {number}: The amount of time (in milliseconds) that commands are allowed to run. Example: `{ "ms": 2000 }`	A JSON object wi Key / Ty sessionId / str status / int value / ob

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POST v1/session/:sessionId/timeouts/press_wait

Method Type	Path	JSON Parameters	Return Value

POST	session/:sessionId/timeouts/press_wait	ms - {number}: The amount of time (in milliseconds) between keypress commands.
		
		Example:
		
		{
		"ms": 2000
		}
			A JSON object with
			
			Key: sessionId, Type: strin
			Key: status, Type: int
			Key: value, Type: objec

POST v1/session/:sessionId/elements

Method Type	Path	JSON Parameters	Return Value
POST	session/:sessionId/elements	An array of the following objects, which can be used to located an element: using - {string}: The locator strategy to use. This may be one of the following values: • text: Returns an element whose text matches the search value. • attr: Returns an element whose specified attributes matches the search value. • tag: Returns an element whose tag name matches the search value. attribute - {string}: The attribute name (used only for "attr" strategy) value - {string}: The search target. Example:	A WebElement JSON object elements. This object has th Key sessionId status value value.XMLName value.XMLName.Local value.XMLName.Space value.Attr value.Attr[i].Name value.Attr[i].Name.Local value.Attr[i].Name.Space value.Attr[i].Value value.Nodes

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Method type	Path	JSON Parameters	Return Value

		<pre>{
		"elementData": [{
		"using": "tag",
		"value": "Label"
		},
		{
		"using": "text",
		"value": "series"
		},
		{
		"using": "attr",
		"attribute": "index"
		"value": "0"
		}
		]
		"parentData": [{
		"using": "tag",
		"value": "Grid"
		}
		]
		}</pre>	

POST v1/session/:sessionId/element

Method Type	Path	JSON Parameters	Return Value
POST	session/:sessionId/element	An elementData array and optional parentData array with the following objects that can be used to locate an element: using - {string}: The locator strategy to use. This may be one of the following values: • text: Returns an element whose text matches the search value. • attr: Returns an element whose specified attributes matches the search value. • tag: Returns an element whose tag name matches the search value. attribute - {string}: The attribute name (used only for "attr" strategy) value - {string}: The search target. Example:	A WebElement JSON object r element. This object has the

Key	T
sessionId	s
status	ir
value	o
value.XMLName	o
value.XMLName.Local	s
value.XMLName.Space	s
value.Attr	a
value.Attr.Name	o
value.Attr.Name.Local	s
value.Attr.Name.Space	s
value.Attr.Value	s
value.Nodes	a

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Method type	Path	JSON Parameters	Return Value

		{
		"elementData": [{
		"using": "tag",
		"value": "Label"
		},
		{
		"using": "text",
		"value": "series"
		},
		{
		"using": "attr",
		"attribute": "index"
		"value": "0"
		}
		]
		"parentData": [{
		"using": "tag",
		"value": "Grid"
		}
		]
		}	

GET v1/session/:sessionId/element/active

Method Type	Path	Return Value
GET	session/:sessionId/element/active	A JSON object with the element that currently l This object has the following fields:

Key	Type	Descript
sessionId	string	The adv of the d
status	int	A status summar result of commar
value	object	
value.XMLName	object	
value.XMLName.Local	string	The nam element
value.XMLName.Space	string	The nam identifie element
value.Attr	array	
value.Attr[i].Name	object	
value.Attr[i].Name.Local	string	The nam attribut
value.Attr[i].Name.Space	string	The nam identifie attribut
value.Attr[i].Value	string	The valu attribut
value.Nodes	array	The chil of the re item.

GET v1/session/:sessionId/source

Method Type	Path	Return Value	Descript
GET	session/:sessionId/source	A JSON object with the current page source. This object has the following fields: Key | Type | Description sessionId | string | The advertising ID of the device. status | int | A status code summarizing the result of the command. value | string | A base64 string that can be decoded to XML.	Retrieve the curr page source.

GET v1/session/:sessionId/apps

Method Type	Path	Return Value	Des
GET	session/:sessionId/apps	A JSON object with an array of installed apps. This object has the following fields: Key | Type | Description sessionId | string | The advertising ID of the device status | int | A status code summarizing the result of the command. value | array | value[i].Title | string | The title of the app. value[i].ID | string | The ID of the app. value[i].Version | string | The build version of the app. value[i].Subtype | string | "ndka"/"rsga" value[i].Type | string | "menu"/"appl"	Ret app inst dev

GET v1/session/:sessionId/current_app

Method Type	Path	Return Value
GET	session/:sessionId/current_app	A JSON object with the app currently loaded on the device. This object has the following fields:

Key	Type	Description
sessionId	string	The advertising ID of the device
status	int	A status code summarizing the result of the command.
value	array	
value[i].Title	string	The title of the app.
value[i].ID	string	The ID of the app.
value[i].Version	string	The build version of the app.
value[i].Subtype	string	"ndka"/"rsga"
value[i].Type	string	"menu"/"appl"

GET v1/session/:sessionId/player

Method Type	Path	Return Value
GET	session/:sessionId/player	A JSON object with the information about the Roku medi object has the following fields:

Key	Type	Des
sessionId	string	The of t
status	int	A st sum resu com
value	object	
value.error	string	Indi the play no e this
value.state	string	Indi curr stat "pau and
value.format	object	The eler the attr *cap drm*
value.format.audio	string	The com met "aac on.)
value.format.caption	string	The forr for valu "nor no c

Method type	Path	Return Value

Key	Type	Des
value.format.container	string	The form exa
value.format.drm	string	The If no use "nor
value.format.video	string	The curr vide ("mp exa
value.format.res	string	The the play stre ("12 exa
value.buffering	object	The eler the attr *max*
value.buffering.current	string	The buff kbp
value.buffering.max	string	The pos spe
value.buffering.target	string	The buff kbp
value.newStream	object	The eler the attr

Method Type	Path	Return Value

Key	Type	Des
value.newStream.speed	string	The play bps
value.position	string	The curr the exp elap ms) of s tim the
value.duration	string	The vide (in s bec whe beg cha is d such eve
value.isLive	string	A fla whe beir live
value.runtime	string	The vide (in s

Method type	Path	Return Value

Key	Type	Des
value.streamSegment	object	The stre attr Info the that stre only seg tran DAS This con follo attr *med seg time*
value.streamSegment.bitrate	string	The vide bps
value.streamSegment.mediaSequence	string	The seq seg vide
value.streamSegment.segmentType	string	The the may follo "auc "cap
value.streamSegment.time	string	The time

Testing production apps

To test production apps with the Roku Web Driver APIs, package the app (/docs/developer-program/publishing/packaging-channels.md#rekeying) on your Roku device using the same Roku developer account linked to the production version of the app.

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